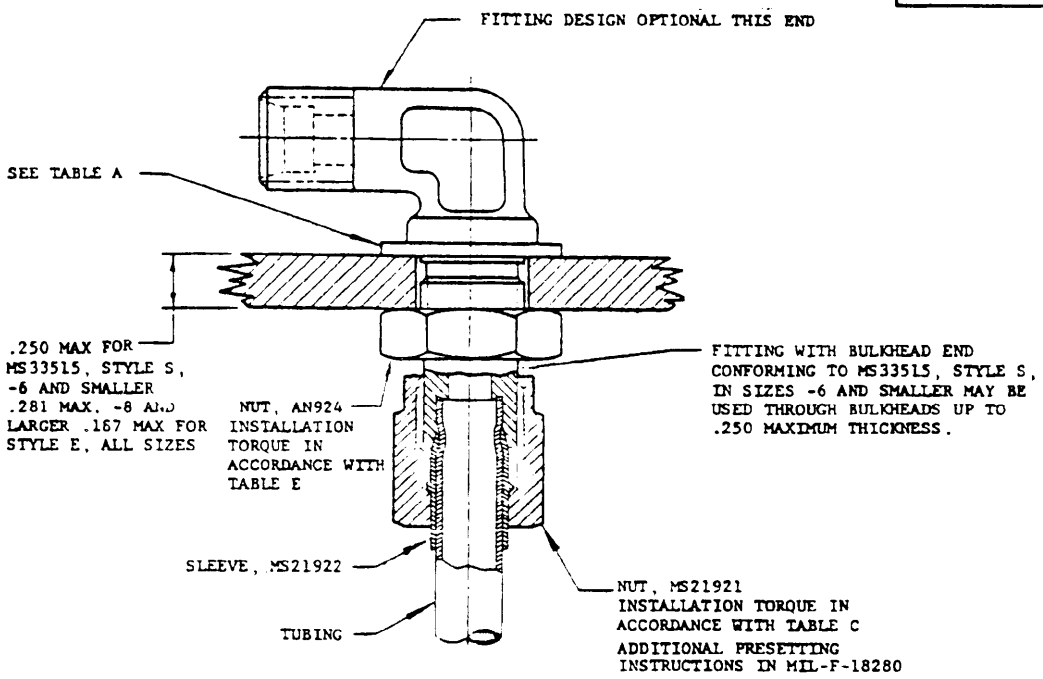
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BULKHEAD FLARELESS FITTING INSTALLATION

★ BULKHEAD HOLE SIZES AND WASHER THICKNESS

TABLE A

FITTING SIZE	BULKHEAD HOLE DIA (REF)	AN960 WASHER THICKNESS (BULKHEAD .187 THICK OR LESS)	FITTING SIZE	BULKHEAD HOLE DIA (REF)	AN960 WASHER THICKNESS (BULKHEAD .187 THICK OR LESS)
-2	.318/.338	.062	-10	.880/ .900	.094
-3	.380/.400		-12	1.068/1.088	
-4	.443/.463		-16	1.318/1.338	
-5	.505/.525		-20	1.630/1.650	
-6	.568/.588		-24	1.880/1.900	
-8	.755/.775	.094	-32	2.505/2.525	

★ NOTE: FOR HOLE SIZES SHOWN (BASED ON AN960ID), NO WASHER REQUIRED WITH MS33515, STYLE S. FOR BULKHEADS THICKER THAN .187, OR WITH MS33515, STYLE E FOR ALL THICKNESSES.

INSTALLATION INSTRUCTIONS

1. HAND PRESETTING OF MS21922 SLEEVE:

   (a) BOTTOM THE TUBE END FIRMLY IN THE SEAT OF THE PRESET TOOL OR FITTING.
   (b) TIGHTEN THE MS21921 NUT 1 1/6 TURNS PAST THE POINT OF RING GRIP ON ALL SIZES AND MATERIALS FOR ALL TYPES OF FITTINGS OR TUBES.

2. RESETTING (INSTALLATION)

   TIGHTEN THE MS21921 NUT IN ACCORDANCE WITH TABLE C.

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NO.

(B) ENTIRE STD REVISED

APPROVED 2 MAY 56 REVISED (A) 28 DEC 62 (B) 20 MAR 75

P A NAVY - AS Other Cust ARMY - AV USAF - 82	TITLE FITTINGS, INSTALLATION OF FLARELESS TUBE, STRAIGHT-THREADED CONNECTORS	MILITARY STANDARD MS 33566
PROCUREMENT SPECIFICATION NONE	SUPERSEDES MS33566(ASG)	SHEET 1 OF 7

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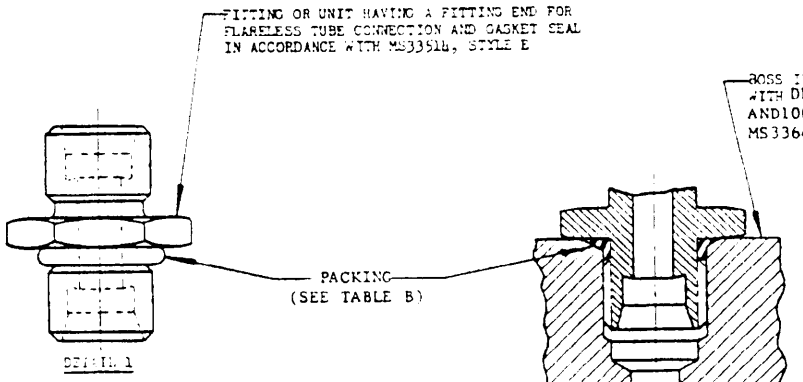
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NONPOSITIONING OR PLUG TYPE FITTING INSTALLATION SUITABLE FOR NOMINAL OPERATING PRESSURES UP TO AND INCLUDING 3,000 PSI

PROCEDURE FOR BOSS INSTALLATION OF MS33514 FITTING END, STYLE E.

1. LUBRICATE THE PACKING IN APPROPRIATE LIQUID (SEE TABLE B).
2. INSTALL PACKING ON THE FITTING AS SHOWN IN DETAIL 1.
3. SCREW THE FITTING ASSEMBLY INTO THE BOSS USING INSTALLATION TORQUE IN ACCORDANCE WITH TABLE D.

TABLE B

PACKING AND LUBRICANT SELECTION		
APPLICATION	DOCUMENT NUMBER	APPROPRIATE LUBRICATION
HYDRAULIC (PETROLEUM BASE)	MS28778,MIL-R-83248/1,MIL-R-83248/2,MS3393	SYSTEM FLUID
HYDRAULIC (PHOSPHATE ESTER)	MS1612	SYSTEM FLUID
PNEUMATIC	MS9385	MIL-G-4343
ENGINE LUBRICANT (PETROLEUM BASE)	MIL-R-83248/1	SYSTEM FLUID
ENGINE LUBRICANT (SYNTHETIC BASE)		
ENGINE FUEL (GASOLINE; JP-3,-4,-5, -6; RJ-1; RP-1; REF-2)		
OXYGEN	MS9385	NONE
VACUUM	MIL-R-83248/2	NONE
COOLANT (ETHYLENE GLYCOL)	MS28778	NONE
COOLANT (SILICATE ESTERS)		

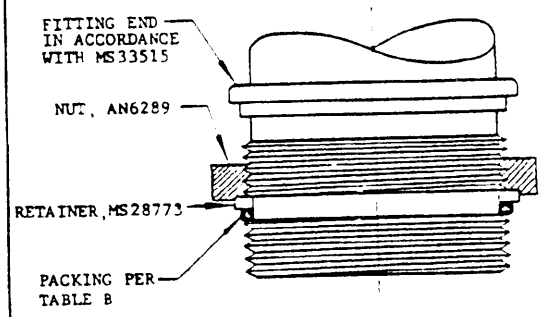
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P A NAVY - AS Other Cust ARMY - AV USAF - 82	TITLE FITTINGS, INSTALLATION OF FLARELESS TUBE, STRAIGHT-THREADED CONNECTORS	MILITARY STANDARD MS33566
PROCUREMENT SPECIFICATION NONE	SUPERSEDES MS33566(ASG)	SHEET 2 OF 7

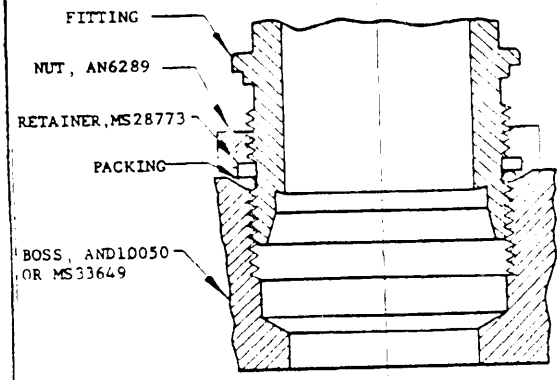
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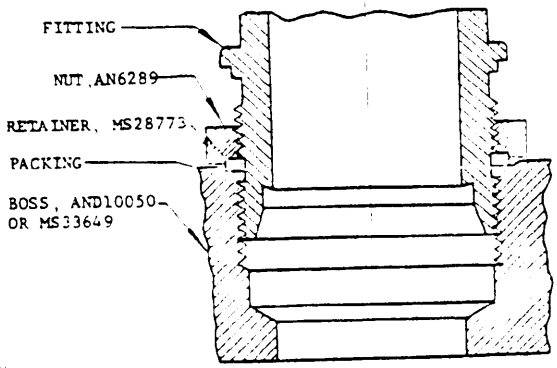
COAT MALE THREADS OF FITTING, RETAINER, AND PACKING SPARINGLY WITH LUBRICANT PER TABLE B AND ASSEMBLE AS SHOWN. WORK THE RETAINER MS28773 INTO THE COUNTERBORE OF THE NUT, AN6289; THEN TURN THE NUT, AN6289, DOWN UNTIL THE PACKING IS PUSHED FIRMLY AGAINST THE LOWER THREADED SECTION OF THE FITTING.

STEP 1



INSTALL THE FITTING INTO THE BOSS AND, AT THE SAME TIME, KEEP THE NUT, AN6289, TURNING WITH THE FITTING UNTIL THE PACKING CONTACTS THE BOSS. THIS POINT CAN BE DETERMINED BY A SUDDEN INCREASE IN TORQUE. WITH THE FITTING IN THIS POSITION, PUT A WRENCH ON THE NUT, AN6289, TO PREVENT ITS TURNING AND, AT THE SAME TIME, TURN THE FITTING IN 1 1/2 TURNS. POSITION FITTING BY TURNING IN NOT MORE THAN ONE ADDITIONAL TURN.

STEP 2



HOLD FITTING AND TURN NUT, AN6289, DOWN TIGHTLY AGAINST BOSS USING INSTALLATION TORQUE IN ACCORDANCE WITH TABLE D.

STEP 3

POSITIONING TYPE UNIVERSAL FITTING INSTALLATION SUITABLE FOR NOMINAL OPERATING PRESSURES UP TO AND INCLUDING 3,000 PSI

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P A NAVY - AS	TITLE	MILITARY STANDARD
Other Cust ARMY - AV USAF - 82	FITTINGS INSTALLATION OF FLARELESS TYPE, STRAIGHT-THREADED CONNECTORS	MS33566
PROCUREMENT SPECIFICATION NONE	SUPERSEDES MS33566(ASG)	SHEET 3 OF -

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TABLE C
TORQUE VALUES FOR MS21921 COUPLING NUT
BASED ON LUBRICATION WITH MIL-H-5606 AND AS NOTED ~~HEREIN~~

TUBE SIZE	*CODE	ALUM. (6061-T6) TUBING WALL THICKNESS (IN.)		TORQUE (LB. IN.) MIN.	MAX.
		HEAVY	THIN		
-2					
-2					
-2	A, B			75	85
-2					
-3	C	.028		95	105
-3					
-3	A, B			95	105
-3					
-4	C	.035		135	145
-4	D		.028	120	135
-4	A, B			135	145
-4					
-5					
-5					
-5	A, B			175 DRY	195 DRY
-5					
-6	C	.049		215	245
-6	D		.028	190	215
-6	A, B			215	245
-6					
-8	C	.065		470	510
-8	D		.035	340	380
-8	D		.049	340	380
-8	A, B			470	510
-10	C	.083		620	680
-10	D		.035	380	450
-10	D		.049	380	450
-10	A, B			620	680
-12	C	.095		855	945
-12	D		.049	400	470
-12	D		.065	400	470
-12	A, B			855	945

CONT.

APPROVED 2 MAY 56 REVISED (A) 24 DEC 62 (B) FOR C SS SF ETS 1 THRU 7

P.A.
NAVY - AS
Other Cust
ARMY - AV
USAF - 82

TITLE
FITTINGS
INSTALLATION OF FLARELESS TUBE, STRAIGHT-THREADED CONNECTORS

MILITARY STANDARD
MS 33566

PROCUREMENT SPECIFICATION
NONE

SUPERSEDES
MS33566(ASG)

SHEET 4 OF -

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TABLE C (CONT'D)

TUBE SIZE	*CODE	ALUM. (6061-T6) TUBING WALL THICKNESS (IN.)	TORQUE (LB. IN.) MIN.	MAX.
-16	C	.065	1140	1260
-16	D	.049	750	850
-16	A, B		1140	1260
-20				
-20	D	.049	1000	1100
-20	A, B		1520	1680
-20				
-24				
-24				
-24	A, B		1900	2100
-24				
-32				
-32				
-32	A, B		2660	2940
-32				

* CODE

A STEEL OR CRES NUT TUBE OR FITTING

B ALUMINUM FITTINGS OR NUTS USED WITH CRES TUBING

C ALUMINUM FITTINGS OR NUTS USED WITH HEAVY WALL ALUMINUM TUBING (6061-T6)

D ALUMINUM FITTINGS OR NUTS USED WITH THIN WALL ALUMINUM TUBING (6061-T6)

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P.A. NAVY - AS Other Cust ARMY - AV USAF - 82	TITLE FITTINGS INSTALLATION OF FLARELESS TUBE STRAIGHT THREADED CONNECTORS	MILITARY STANDARD MS33566
PROCUREMENT SPECIFICATION NONE	SUPERSEDES MS 33566 (ASG)	SHEET 5 OF 7

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TABLE D

TORQUE VALUES FOR BOSS INSTALLATIONS USING PACKINGS (LB. IN.) (FITTINGS PER MS33514 STYLE E, AN6289 NUT USED WITH FITTINGS PER MS33515, STYLE E OR S, ALL MATERIALS, BASED ON LUBRICATION WITH MIL-H-5606 AND AS NOTED HEREIN.)

SIZE	TUBE OD (IN.) (REF)	THREAD	TORQUE MIN	TORQUE MAX
-2	1/8 (.125)	5/16-24	50	55
-3	3/16 (.187)	3/8 -24	65	75
-4	1/4 (..250)	7/16-20	95	105
-5	5/16 (.312)	1/2 -20	125 DRY	135 DRY
-6	3/8 (.375)	9/16-18	155	165
-8	1/2 (.500)	3/4 -16	280	305
-10	5/8 (.625)	7/8 -14	380	405
-12	3/4 (.750)	1 1/16-12	550	600
-16	1 (1.000)	1 5/16-12	800	900
-20	1 1/4 (1.250)	1 5/8 -12	900	1000
-24	1 1/2 (1.500)	1 7/8 -12	900	1000
-32	2 (2.000)	2 1/2 -12	900	1000

TABLE E

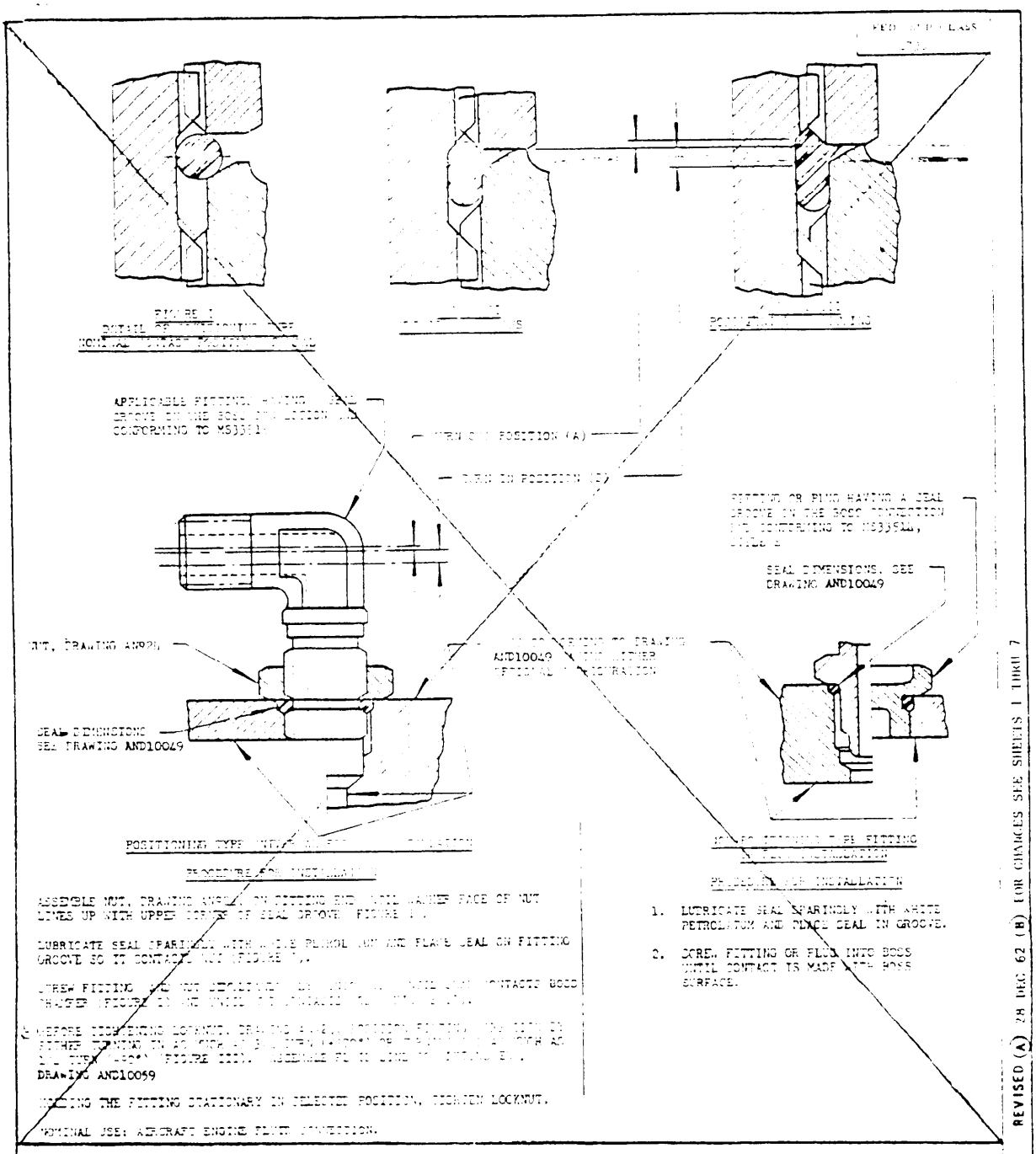
TORQUE VALUES FOR AN924 JAM NUT USED ON FITTINGS PER MS33515 STYLE E OR S IN BULKHEAD INSTALLATIONS (LB. IN.)

SIZE	TUBE OD (IN.) (REF)	THREAD	TORQUE MIN	TORQUE MAX
-2	1/8 (.125)	5/16-24	35	50
-3	3/16 (.187)	3/8 -24	65	80
-4	1/4 (.250)	7/16-20	85	105
-5	5/16 (.312)	1/2 -20	105	125
-6	3/8 (.375)	9/16-18	120	150
-8	1/2 (.500)	3/4 -16	240	280
-10	5/8 (.625)	7/8 -14	320	380
-12	3/4 (.750)	1 1/16-12	500	600
-16	1 (1.000)	1 5/16-12	720	880
-20	1 1/4 (1.250)	1 5/8 -12	960	1200
-24	1 1/2 (1.500)	1 7/8 -12	1200	1440
-32	2 (2.000)	2 1/2 -12	1400	1500

NOTE: FITTING SHALL BE RESTRAINED FROM TURNING WHEN TORQUING TO VALUES GIVEN IN TABLES C AND E.

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P A	TITLE	MILITARY STANDARD
NAVY - AS Other Cust ARMY - AV USAF - 82	FITTINGS, INSTALLATION OF FLARELESS TUBE, MECHANICALLY-FORMED CONNECTORS	MS33566
PROCUREMENT SPECIFICATION SUPERSEDES NONE		SHEET 7 OF 7

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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